THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · NCERT-TYPE UNIT TEST SET 3

Chapters 8–11: **Introduction to Trigonometry, Some Applications of Trigonometry, Circles, Areas Related to Circles** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A tangent PQ at a point P of a circle of radius 5 cm meets a line through the centre O at a point Q so that $OQ = 12$ cm. The length of PQ is: 1

(a) 12 cm **(b)** 8.5 cm **(c)** 13 cm **(d)** $\sqrt{119}$ cm

2. A circular field has radius 21 m. Fencing costs ₹ 5 per metre. The cost (in ₹) of fencing the field once is: 1

(a) ₹ 132 **(b)** ₹ 6930 **(c)** ₹ 660 **(d)** ₹ 330

3. If $\cos A = \frac{8}{17}$, then $\tan A =$ 1

(a) $\frac{17}{8}$ **(b)** $\frac{8}{15}$ **(c)** $\frac{15}{17}$ **(d)** $\frac{15}{8}$

4. A ladder 16 m long leans against a vertical wall and makes an angle of 60° with the level ground. The height reached on the wall is: 1

(a) 8 m **(b)** $16\sqrt{3}$ m **(c)** $8\sqrt{3}$ m **(d)** $4\sqrt{3}$ m

5. A point P is 35 cm from the centre O of a circle of radius 21 cm. The length of the tangent drawn from P to the circle is: 1

(a) 14 cm **(b)** 784 cm **(c)** 56 cm **(d)** 28 cm

6. A wheel has radius 35 cm. The number of complete revolutions it makes to cover 110 m is: 1

(a) 50 **(b)** 60 **(c)** 25 **(d)** 100

7. The value of $\tan^2 60^\circ - \tan^2 30^\circ$ is: 1

(a) $\frac{11}{3}$ **(b)** $\frac{16}{3}$ **(c)** 3 **(d)** $\frac{8}{3}$

8. A boy whose eyes are 3 m above the ground stands 15 m away from a vertical building. The angle of elevation of the top of the building from his eyes is 45°. The height of the building is: 1

(a) 19 m **(b)** 18 m **(c)** $18\sqrt{3}$ m **(d)** 15 m

9. A quadrilateral $ABCD$ is drawn about a circle so that all four sides touch the circle. $AB = 12$ cm, $BC = 10$ cm and $CD = 8$ cm. The length of AD is: 1

(a) 6 cm **(b)** 10 cm **(c)** 30 cm **(d)** 14 cm

10. The radii of two circles are 8 cm and 6 cm. The radius of the circle whose circumference equals the sum of their circumferences is: 1

(a) 28 cm **(b)** 10 cm **(c)** 7 cm **(d)** 14 cm

11. $\tan 35^\circ \cdot \tan 55^\circ =$ 1

(a) 1 **(b)** 35 **(c)** 0 **(d)** 2

12. The angle of elevation of the top of a vertical tower from a point on level ground is 30°. After walking 30 m towards the foot of the tower along a straight line, the angle of elevation becomes 60°. The height of the tower is: 1

(a) 15 m **(b)** $30\sqrt{3}$ m **(c)** $10\sqrt{3}$ m **(d)** $15\sqrt{3}$ m

13. PA and PB are tangents from an external point P to a circle with centre O, touching it at A and B. $\angle APB = 70^\circ$. The $\angle OAB$ is: 1

(a) 70° **(b)** 55° **(c)** 35° **(d)** 20°

14. The radii of two circles are 5 cm and 12 cm. The radius of the circle whose area equals the sum of their areas is: 1

(a) 13 cm **(b)** 7 cm **(c)** 17 cm **(d)** 15 cm

15. The value of $\sin 22^\circ / \cos 68^\circ$ is: 1

(a) 22 **(b)** 0 **(c)** 2 **(d)** 1

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. A quadrilateral $ABCD$ is drawn about a circle so that all four sides touch the circle. $AB = 13$ cm, $BC = 9$ cm and $CD = 8$ cm. Find the length of AD. 2

17. A horse is tied at a corner of a large square field. The rope is first 7 m long and then lengthened to 14 m. Find the increase in the area it can graze. 2

18. If $\tan A = \frac{12}{5}$, find $\sec A$. 2

19. A storm breaks a tree at a height of 8 m above the ground. The broken part bends over, its top touches the ground and the broken part makes an angle of 30° with the ground. Find the distance from the foot of the tree to the point where its top touches the ground. 2

20. Tangents QA and QB are drawn from an external point Q to a circle, and $QA = (5x + 1)$ cm and $QB = (3x + 9)$ cm. Find the length of QA. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. The incircle of $\triangle ABC$ touches BC, CA and AB at D, E and F. $AB = 14$ cm, $BC = 10$ cm and $CA = 12$ cm. Find the lengths AF, BD and CE (in this order). 3

22. A chord of a circle of radius 14 cm subtends a right angle at the centre. Find the area of the minor segment. 3

23. In $\triangle ABC$, right-angled at B, $AB = 20$ cm and $BC = 21$ cm. Find $\sin A$, $\cos A$ and $\tan A$. 3

24. From the top of a building 8 m high, the angle of elevation of the top of a tower is 60° and the angle of depression of its foot is 45°. Find the height of the tower. 3

OR

In $\triangle ABC$, right-angled at B, $AB = 24$ cm and $BC = 32$ cm. A circle is inscribed in the triangle. Find the radius of the inscribed circle. 3

25. A sector of a circle of radius 14 cm has central angle 180°. Find the perimeter of the sector. 3

OR

Prove that: $\tan\theta + \cot\theta = \sec\theta\,\csc\theta$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Prove that: $(\csc\theta - \cot\theta)^2 = \dfrac{1-\cos\theta}{1+\cos\theta}$. 5

OR

Two vertical poles of equal height stand on opposite sides of a road 120 m wide. From a point on the road between them, the angles of elevation of the tops of the poles are 60° and 30°. Find the common height of the poles and the distances of the point from the poles (nearer pole first). 5

27. Prove that the lengths of the tangents drawn from an external point to a circle are equal. 5

OR

A chord of a circle of radius 21 cm subtends an angle of 60° at the centre. (Use $\pi = \frac{22}{7}$ and $\sqrt{3} = 1.73$; answer to two decimal places.) Find the area of the minor segment. 5

END OF QUESTION PAPER — Total: 50 Marks

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